AI Advisor: Multilingual Retrieval for a RAG Assistant

Alfredo Alea · Data / AI Engineer · Own prototype, 2026 · Rust (Axum), Qdrant, fastembed, OpenRouter, SSE · Python benchmark · Code: github.com/cursorcanis/portfolio-ai-machine-learning-engineer

BUSINESS QUESTION

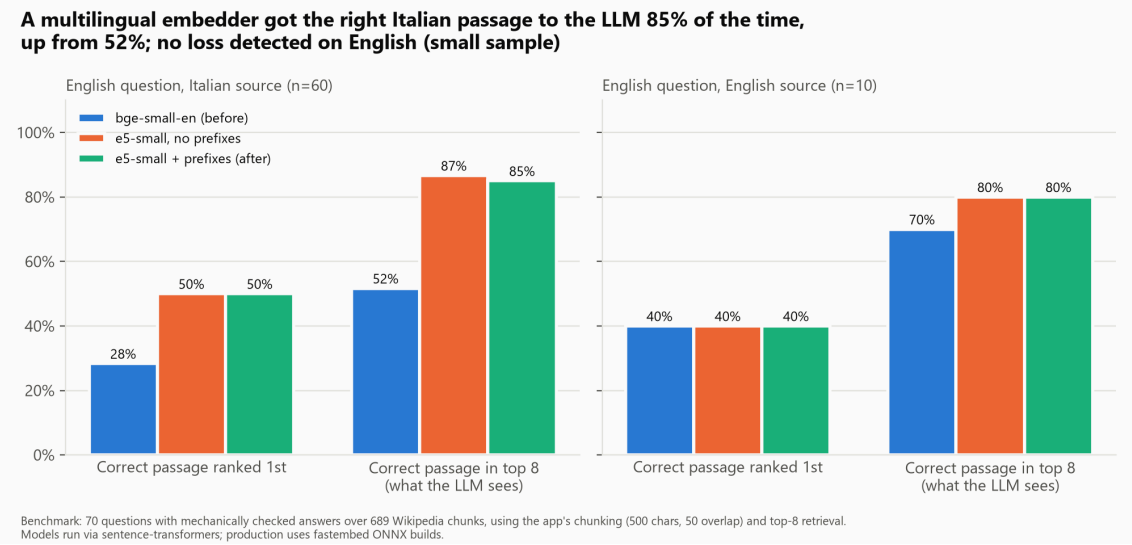
Answers drawn from Italian sources were noticeably worse than answers from English ones. Should search keep depending on a rate-limited service that translates Italian transcripts into English, or switch to a model that reads both languages natively, as long as English quality doesn't suffer?

METHOD

- Replaced the English-only embedder (bge-small-en-v1.5) with multilingual-e5-small (same 384 dimensions) and rebuilt the Qdrant collection.
- Enforced E5's `query:`/`passage:` prefixes inside the embedder after confirming the library doesn't add them.
- Benchmarked 70 questions over 689 Wikipedia chunks (Italian and English), using the app's chunking (500 characters, 50 overlap) and top-8 retrieval. Retrieval hits were scored by exact match on the gold answer text.

KEY FINDINGS

English question about an…	Before	After
Italian source: correct passage in top 8 (what the LLM sees)	52%	**85%** (21 questions won, 1 lost)
Italian source: correct passage ranked 1st	28%	**50%** (95% CI of gain +7 to +37 pts)
Italian source, excluding word-overlap questions (n=53), top 8	47%	**83%**
English source (n=10), top 8	70%	80% (no loss detected, small sample)



RECOMMENDATIONS

1. Keep the multilingual embedder. Retrieval no longer depends on YouTube's rate-limited translation endpoint. Translation remains only as an optional step for uploaded documents, so answers can come back in English.
2. Re-run the benchmark before every model change.
3. Next: Italian-language questions, a 50+ question English set, a run on the production ONNX builds, and a cross-encoder reranker for top-1.

VERIFICATION

- I discarded my first evidence (similarity score 0.62 → 0.81) because scores aren't comparable across models.
- The original 32 questions are kept inside the 70. I expanded the set because the first interval crossed zero, and I'm disclosing that the stopping rule wasn't fixed in advance.
- The gain survives removing the questions that could be solved by word overlap, and the prefixes made no measurable difference.

Also shipped: incremental backend migration from Next.js/Node to Rust (strangler fig) · 34 unit tests · 35 MB stripped release binary (thin LTO), down from a ~99 MB debug build · token streaming over SSE.